

Exponentials and logarithms - Set 1 - Solutions

Specification MM5.1-MM5.3.

Answer key

Question	Answer	Topic
1	A	Exponentials and logarithms
2	B	Exponentials and logarithms
3	F	Exponentials and logarithms
4	B	Exponentials and logarithms
5	C	Exponentials and logarithms
6	C	Exponentials and logarithms
7	D	Exponentials and logarithms
8	A	Exponentials and logarithms
9	C	Exponentials and logarithms
10	B	Exponentials and logarithms
11	F	Exponentials and logarithms
12	B	Exponentials and logarithms

Worked solutions

Question 1 LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

Answer: A

$$\log_2 \left(\frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

Question 2 LOG-02 · Exponentials and logarithms · spec MM5.3

Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

Answer: B

$$2x = \log_3 4 = 2\log_3 2, \text{ so } x = \log_3 2.$$

Question 3 LOG-03 · Exponentials and logarithms · spec MM5.3

Solve $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$. Find the sum of the solutions.

Answer: F

$$\text{Let } y = 2^x: y^2 - 12y + 32 = (y - 4)(y - 8) = 0, \text{ so } 2^x = 4 \text{ or } 8: x = 2 \text{ or } 3. \text{ Sum } 5.$$

Question 4 LOG-04 · Exponentials and logarithms · spec MM5.2

Given that $\log_a x = 3$ and $\log_a y = 5$, find $\log_a \left(\frac{x^2 \sqrt{y}}{a} \right)$.

Answer: B

$$2\log_a x + \frac{1}{2}\log_a y - \log_a a = 6 + \frac{5}{2} - 1 = \frac{15}{2}.$$

Question 5 LOG-05 · Exponentials and logarithms · spec MM5.2

Solve $\log_3 (x + 6) + \log_3 (x - 2) = 2$.

Answer: C

$$(x + 6)(x - 2) = 9 \Rightarrow x^2 + 4x - 21 = 0 \Rightarrow (x + 7)(x - 3) = 0. x = -7 \text{ makes both logs undefined, so } x = 3 \text{ only.}$$

Question 6 LOG-07 · Exponentials and logarithms · spec MM5.2

Which of the following statements is/are true for all positive real x and y ?

I $\log_{10} (x + y) = \log_{10} x + \log_{10} y$

II $\log_{10} (xy)^2 = 2\log_{10} x + 2\log_{10} y$

III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

Answer: C

II is correct: $\log(xy)^2 = 2\log(xy) = 2\log x + 2\log y$. I and III confuse the laws: log of a sum has no simple form, and a difference of logs is log of a quotient, not a quotient of logs.

Question 7 LOG-09 · Exponentials and logarithms · spec MM5.2

Find the value of $\frac{\log_5 27}{\log_5 3}$.

Answer: D

$$\log_5 27 = \log_5 3^3 = 3\log_5 3, \text{ so the quotient is } 3.$$

Question 8 LOG-10 · Exponentials and logarithms · spec MM5.3

How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?

Answer: A

$$y = 2^x: y^2 + 2y - 8 = (y + 4)(y - 2) = 0. y = -4 \text{ is impossible, } y = 2 \text{ gives } x = 1. \text{ One solution.}$$

Question 9 LOG-11 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

Answer: C

$$\frac{3}{2}\log_2 x = 6 \Rightarrow \log_2 x = 4 \Rightarrow x = 16.$$

Question 10 LOG-12 · Exponentials and logarithms · spec MM5.1

The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .

Answer: B

$$\frac{a^4}{a^2} = a^2 = \frac{108}{12} = 9, \text{ so } a = 3 \text{ (as } a > 0).$$

Question 11 LOG-13 · Exponentials and logarithms · spec MM5.2

Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

Answer: F

$$0.75 = \frac{3}{4}, \text{ so } \log 0.75 = \log 3 - \log 4 = q - 2p.$$

Question 12 LOG-15 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 (x^2 - 4) - \log_2 (x - 2) = 3$.

Answer: B

$$\text{For } x > 2: \log_2 \frac{(x-2)(x+2)}{x-2} = \log_2 (x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6.$$